Antoine Bouyeure

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Experience

Dec 2021–Present **Postdoctoral Researcher**, *Ruhr University Bochum*, Bochum, Germany
Supervisor: Nikolai Axmacher

Oct 2017–Oct 2021 **PhD Candidate**, *University of Paris*, Paris, France
Supervisor: Marion Noulhiane

Nov 2019–Feb 2020 **Visiting PhD Candidate**, *Donders Institute, Radboud University*, Nijmegen, Netherlands

Jan 2017–Jun 2017 **Research Intern**, *Neurospin, CEA*, Saclay, France

Feb 2017 **Visiting Research Intern**, *Radboud University*, Nijmegen, Netherlands

Jun 2016–Oct 2016 **Research Intern**, *University of Paris*, Paris, France

Sep 2014–Sep 2015 **Full-time Research Assistant in Epistemology**, *University of Paris*, Paris, France

Education

2017–2021 **PhD in Neurosciences**, *University of Paris*, Paris, France

2015–2017 **MSc in Cognitive Neurosciences**, *Sorbonne University*, Paris, France

2014–2015 **First year of MSc in Genetics**, *University of Paris*, Paris, France

2011–2014 **BS in Psychology**, *University of Paris*, Paris, France

2009–2014 **BS in Philosophy and MA in Philosophy of Sciences**, *Sorbonne University*, Paris, France

Publications and Communications

Research Papers

2025 **Bouyeure, A.**, Pacheco, D., Fellner, M. C., Jacob, G., Kobelt, M., Rose, J., & Axmacher, N. *Distinct representational properties of cues and contexts shape fear learning and extinction.* **eLife**, *In press.*

2025 Pacheco-Estefan, D., **Bouyeure, A.**, Jacobs, G., Fellner, M. C., Lehongre, K., Lambrecq, V., ... & Axmacher, N. *Representational dynamics during extinction of fear memories in the human brain.* **Nature Human Behavior**, 1-20.

2025 Ahmadi, K., Swegle, S., Kashyap, S., **Bouyeure, A.**, Bandettini, P., Axmacher, N., & Huber, L. *Blood volume sensitive laminar fMRI with VASO in human hippocampus: Capabilities and biophysical challenges at clinical 7T scanners.* **bioRxiv**, 2025-08.

2025 **Bouyeure, A.***, Pfaffenrot, V.*, Gomes, C.A., Kashyap, S., Axmacher, N., Norris, D.G. *Characterizing BOLD activation patterns in the human hippocampus with laminar fMRI.* **Imaging Neuroscience**, 3, 005032.

2025 Allinger, J., Noulhiane, M., Féménias, D., Louvet, B., Clua, E., **Bouyeure, A.**, Lemaître, F. *Risk profiles of elite breath-hold divers.* **International Journal of Environmental Health Research**, 35(3), 680-692.

2024 Allinger, J., **Bouyeure, A.**, Noulhiane, M., & Lemaitre, F. *Monitoring the Breath-Hold Training Load during an Ecological Session: A Pilot Study.* **International Journal of Sports Medicine**, 45(11), 837-843.

2024 Pfaffenrot, V., **Bouyeure, A.**, Axmacher, N., & Norris, D. G. *Laminar fMRI of the human hippocampus: Accounting for physiological fluctuations and vasculature.* **Proc Intl Soc Mag Reson Med** (Vol. 32, No. 3128).

2024 Poiret, C., **Bouyeure, A.**, Patil, S., Boniteau, C., Duchesnay, E., Grigis, A., ... & Noulhiane, M. *Attention-gated 3D CapsNet for robust hippocampal segmentation.* **Journal of Medical Imaging**, 11(1), 014003-014003.

2023 Poiret, C., **Bouyeure, A.**, Patil, S., Grigis, A., Duchesnay, E., Faillot, M., ... & Noulhiane, M. *A fast and robust hippocampal subfields segmentation: HSF revealing lifespan volumetric dynamics.* **Frontiers in Neuroinformatics**, 17, 1130845.

2022 **Bouyeure, A.**, Bekha, D., Patil, S., Hertz-Pannier, L., Noulhiane, M. *Maturity of white matter tracts is associated with episodic memory recall during development.* **Cerebral Cortex Communications**, 3(1).

2021 **Bouyeure, A.**, Patil, S., Mauconduit, F., Poiret, C., Isai, D., Noulhiane, M. *Hippocampal subfield volumes and memory discrimination in the developing brain.* **Hippocampus**, 31(11), 1202–1214.

2018 **Bouyeure, A.**, Germanaud, D., Bekha, D., Delattre, V., ... Noulhiane, M. *Three-Dimensional Probabilistic Maps of Mesial Temporal Lobe Structures in Children and Adolescents Brains.* **Frontiers in Neuroanatomy**, (12) 98.

Prep. **Bouyeure, A.**, Poiret, C., Bekha, D., Patil, S., Noulhiane, M. *Functional connectivity correlated of autobiographical memory during childhood.* (In preparation).

Chapters

2021 **Bouyeure, A.**, & Noulhiane, M. *Memory: Normative development of memory systems.* In Martin, Preedy, Rajendram (eds): **Factors affecting neurodevelopment** (pp. 517-527). Academic Press.

2020 **Bouyeure, A.**, Noulhiane, M. *Normal and pathological development of episodic memory: the importance of critical periods.* In Gallagher, Bulteau, Cohen and Michaud (eds): **Neurocognitive Development: Normative Development**. Handbook of clinical neurology (3rd series) (Vol. 173, pp. 201-213), Elsevier.

2019 Grialou, P., Chiron, J., **Bouyeure, A.**, Noulhiane, M., Chiron, P. *[Know thyself: using the Progymnasmata as metacognitive training].* In Chiron, P. (ed): **Practices of Progymnasmata: from Antiquity to the present day.** Editions Rue d'Ulm.

Talks

2025 **Bouyeure, A.**, Axmacher, N. *The neural mechanisms of unwanted trauma memories.* European Congress of Neuropsychopharmacology 2025, Amsterdam, The Netherlands.

2025 **Bouyeure, A.**, Axmacher, N. *Layer-fMRI of the human hippocampus during pattern separation.* Spring Hippocampal Research Conference 2025, Verona, Italy.

Posters

2025 **Bouyeure, A.**, Ahmadi, K., Pfaffenrot, V., Huber, L., Norris, D., Axmacher, N. *Layer-fMRI of the human hippocampus with GRE-BOLD and VASO during autobiographical memory.* GEM 2025, Ruhr-University Bochum, Germany.

2024 **Bouyeure, A.**, Pfaffenrot, V., Norris, D., Axmacher, N. *Layer-fMRI of the human hippocampus during autobiographical memory.* OHBM 2024, Seoul, South Korea.

2024 **Bouyeure, A.**, Pfaffenrot, V., Axmacher, N. *Tracking neural representations of cues and contexts during fear acquisition and extinction.* OHBM 2024, Seoul, South Korea.

2023 **Bouyeure, A.**, Pfaffenrot, V., Axmacher, N. *Tracking neural representations during fear acquisition and extinction.* OHBM 2023, Montreal, Canada.

2021 **Bouyeure, A.**, Champy, S., Poiret, C., Sandesh, P., Noulhiane, M. *Functional connectivity correlates of autobiographical memory in the developing brain.* OHBM 2021, Virtual Conference.

2021 Poiret, C., **Bouyeure, A.**, Patil, S., Duchesnay, E., Grigis, A., Lemaître, F., Noulhiane, M. *Viewpoint Equivariance: Are 3D CapsNets Any Good for Hippocampal Segmentation?.* OHBM 2021, Virtual Conference.

2020 **Bouyeure, A.**, Chauvin, R., Haak, K.V., Beckmann, C.F., Noulhiane, M. *Functional organization of the hippocampus on its anteroposterior axis during childhood.* OHBM 2020, Virtual Conference.

2020 **Bouyeure, A.**, Patil, S., Noulhiane, M. *Hippocampal subfields volume predict memory discrimination performance in the developing brain.* OHBM 2020, Virtual Conference.

2019 **Bouyeure, A.**, Bekha, D., Hertz-Pannier, L., Noulhiane, M. *White matter microstructure is associated to episodic memory performance in the developing brain.* European Brain and Behaviour Society 2019, Prague, Czech Republic.

2018 **Bouyeure, A.**, Bekha, D., Noulhiane, M. *Resting-state functional connectivity is correlated to episodic memory performance in the developing brain.* Society for Neurosciences 2018, San Diego, CA, USA.

2017 **Bouyeure, A.**, Bekha, D., Germanaud, D., Delattre, V., Rivière, D., Mangin, J.F., Chiron, C., Hertz-Pannier, L., Noulhiane, M. *Sulcal morphology of the medial temporal lobe in children and adolescents.* OHBM 2017, Vancouver, Canada.

2017 **Bouyeure, A.**, Bekha, D., Germanaud, D., Delattre, V., Rivière, D., Mangin, J.F., Chiron, C., Hertz-Pannier, L., Noulhiane, M. *Sulcal morphology of the medial temporal lobe in healthy preterm infants.* OHBM 2017, Vancouver, Canada.

Awards

Grant — Mustela Foundation, 3.5k euros.

Fellowship — PhD Fellowship, Ecole Doctorale Cerveau, Cognition, Comportement (3 years).

Teaching

2022–Present — Methods in Neuroimaging (Master's course), Ruhr-University Bochum.

2017–2019 — Psychophysiology (Bachelor's course), University of Paris (144 hours taught total).

2017–2023 — Co-supervision of 8 Master's student internships, University of Paris and Ruhr-University Bochum.

Technical Skills

Languages — Python and R (proficient), Bash, MATLAB, SQL

Statistics — Parametric and non-parametric statistical testing, Machine Learning, Bayesian statistics, Multivariate methods

Tools — AWS, Git, SLURM

Science Outreach

- **Semaine du Cerveau**: Moderator of Jessica Dubois' general public conference during a science outreach event organized by the CEA.
- **Cognitive sciences on the stage**: General public conference in a Parisian theater.
- **Les Savanturiers**: Scientific workshops and activities in an elementary school of the Paris region.
- **Projet Declics**: Workshop about scientific research in a Parisian high school.

Institutional Responsibilities

- **International Graduate Neuroscience School, Ruhr-University Bochum (2024–Present)**: Member of the steering committee of a graduate program.
- **CNRS Research group on Memory (2019–2021)**: Student member of the steering committee of a multidisciplinary CNRS research group.